Report Nº 5
Team Out of Lane
Date 21.05.2025

1 Introduction

This report details the activities carried out to date in the 5th sprint. These activities emphasize the positioning system and the path planning algorithm, as well as improvements in the object detection model.

2 Planned activities

Category	*Activity*	*Responsible*
Hardware improvements	Research hardware additions (headlight)	Alexander P., Carolina N. and Evelyn C.
	Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Testing sensors and actuators for proper capture, processing and response.	Alexander P., Carolina N., Evelyn C. and Pablo F.
Sensing and input package	Define use-cases and test given servers information (localization on map, cars interaction, GPS interaction, interaction with objects)	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Environmental server interaction	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Introduce noise on systems to validate robustness (loss of image, burned image road search, undefined objects and states).	Alexander P., Carolina N., Evelyn C. and Pablo F.
Computer vision package	Intersection detection	Alexander P., Carolina N, Evelyn C. and Pablo F.
	Implement traffic sign detection	Carolina N., Evelyn C. and Pablo F.
	Implement traffic light detection	Evelyn C.
Project Documentation	Architecture revision	Pablo F.
	Architecture diagram	Evelyn C. and Pablo F.
	Prepare traffic sign dataset	Carolina N. and Evelyn C.
	Define decision making (priorities of actions and state flow)	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Define robustness and safety measures	Alexander P., Carolina N, Evelyn C. and Pablo F.
	Project presentation.	Alexander P., Carolina N, Evelyn C. and Pablo F.
	Document the 5th report	Carolina N. and Pablo F.

3 Status of planned activities

Activity	*%*	*Observations*
Research hardware additions (headlight)	100%	The ultrasonic sensors and headlights were removed after extra testing on their usefulness.
Define other necessary sensors, define use-case, integration, preprocessing, noise cancelling	100%	We started using the IMU data and added an opto-isolator to calculate the position of the car based on its trajectory. The positioning system is ready.
Testing sensors and actuators for proper capture, processing and response.	70%	We have tested sensors and actuators. Extra testing is needed.
Define use-cases and test given servers information	20%	We began investigating the data that will be used and potential interactions and behaviors. *(*)*
Environmental server interaction	20%	We began investigating the data that will be used and potential interactions and behaviors.

Activity	*%*	*Observations*
Introduce noise on systems to validate robustness.	10%	We still need to work on making the system more robust.
Intersection detection	70%	We have been working on the detection of intersections and how to proceed when the car is at one (using the GPS system mentioned beforehand).
Implement traffic sign detection	90%	The car is capable of recognizing traffic signs and reacting to most of them. The only signs for which the reaction has not yet been implemented are the *One-way Road* and *No-entry Road signs*.
Implement traffic light detection	90%	We have started working and implementing the traffic light detector. Further testing is needed.
Architecture revision	100%	It was not necessary to modify our architecture.
Architecture diagram	100%	It was not necessary to modify our architecture diagrams.
Prepare traffic sign dataset	90%	The data set is almost ready, but we leave a percentage to improve it if necessary.
Define decision making	90%	We have improved the decision-making process. We have created a node map of our test racetrack to follow a path even in poor visibility scenarios, to improve the decision making.
Define robustness and safety measures	50%	We are working on the overall error handling and recovery of the system.
Project presentation	50%	We have created an Instagram account and several videos and presentations showcasing our work.
Document the 5th report	100%	The report itself.

(*) This activity is planned to last for two sprints to refine the car's reaction.

4 General status of project

The team has made advances in developing and refining the car's algorithm. The car is already capable of following lanes, detecting and reacting to traffic signs, adjusting its speed and/or braking, and following a path based on defined nodes.

5 Upcoming activities

To be continued from the 5th sprint

- Testing sensors and actuators for proper capture, processing and response.
- Define use-cases and test given servers information.
- Testing sensors and actuators for proper capture, processing and response.
- Environmental server interaction.
- Introduce noise on systems to validate robustness (loss of image, burned image road search, undefined objects and states).
- Intersection detection.
- Implement traffic sign detection.
- Implement traffic light detection.
- Prepare traffic sign dataset.
- Define decision making (priorities of actions and state flow).
- Define robustness and safety measures.
- Project presentation.
- Architecture revision
- Architecture diagram.